University of Barishal

Department of Computer Science and Engineering

Course Title: Discrete Mathematics

Course Code: CSE-1203

1st Year 2nd Semester Final Examination

Admission Session: 2020-21

Time: 03 Hours

Marks: 60

N.B.: Answer any **Five** questions out of the followings. All parts of each question must be answered consecutively. Right side of the question shows the maximum marks.

1.a) Define logic and proposition with example. **4**

b) What is bi-implication? Develop truth table of $(p \wedge q) \vee (\neg q \wedge r)$ and $\neg q \vee (\neg p \wedge q)$. **4**

c) How can this English sentence be translated into logical expression? "You cannot ride the roller coaster if you are under 4 feet tall unless you are older than 16 years old." **4**

2.a) Define predicates and quantifiers with examples. **3**

b) What do you mean by "$((p \rightarrow q) \wedge \neg p) \rightarrow \neg q$ is not a tautology"? For each of these arguments, explain with rules of inference are used for each step. **4**

c) Using Mathematical induction method show that **5**

$$\sum_{i=0}^{n-1} 2^i(n-i) = 2^{n+1} - n - 2$$

3.a) Define set with example. What is the power set of {x, y, z, 5} and Cartesian products of {x, y, z} and {a, b, c}. **4**

b) Show that $\forall x\,(P(x) \wedge Q(x))$ and $\forall x\, P(x) \wedge \forall x\, Q(x)$ are logically equivalent. **4**

c) Explain why $(A \times B) \times (C \times D)$ and $A \times (B \times C) \times D$ are not the same. **4**

4.a) Let $p(n) = a_0 + a_1 n + a_2 n^2 + \ldots + a_m n^m$; that is, suppose degree $p(n) = m$, prove that $p(n) = O(n^m)$. **4**

b) Show that if n is an integer greater than 1, then n can be written as a prime or product primes. **4**

c) Analysis the complexity of Quick Sort Algorithm for the following input. **4**

A	C	B	E	D

5.a) Discuss generalized pigeonhole principle and proof this theorem using proof by contradiction. **3**

b) Suppose that there are eight runners in a race. The winner receives a gold medal, the second-place finisher receives a silver medal, and the third-place finisher receives a bronze medal. How many different ways are there to award to these medals, if all possible outcomes of the race can occur and there no ties? **3**

c) How many bit strings of length 10 contain at least three 1s and at least three 0s? **3**

d) A bag contains 10 red marbles, 10 white marbles, and 10 blue marbles. What is the minimum no. of marbles you have to choose randomly from the bag to ensure that we get 4 marbles of same color? **3**

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6.a) Suppose that A and B are events from a sample space S such that $P(A) \neq 0$ and $P(B) \neq 0$. Then proof the following: **5**

$$p(B|A) = \frac{p(A \mid B)\, p(B)}{p(A \mid B)\, p(B) + P(A \mid \bar{B})\, p(\bar{B})}$$

b) Let E1 and E2 be events in the sample space S. Then proof the following: **4**

$$P(E1 \cup E2) = P(E1) + P(E2) - P(E1 \cap E2)$$

c) For each of these relations on the set {1, 2, 3, 4}, decide whether it is reflexive, whether it is symmetric, whether it is antisymmetric, and whether it is transitive with explanation. **3**

i) {(2, 2), (2, 3), (2, 4), (3, 2), (3, 3), (3, 4)}
ii) {(2, 4), (4, 2)}
iii) {(1, 1), (2, 2), (3, 3), (4, 4)}

7.a) Define Isomorphic graph, Complete graph and Bipartite graph with example. **4**

b) Explain Eulerian and Hamiltonian graphs with examples, also draw the graphs of the following: **5**

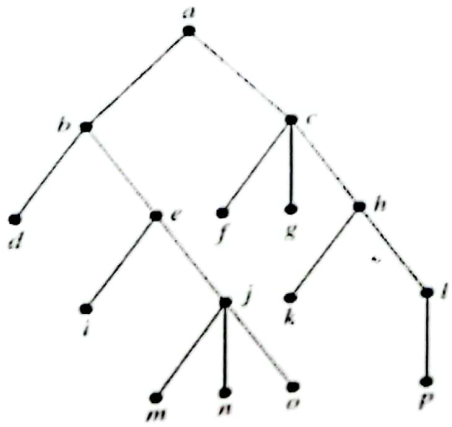
i) Eulerian but not Hamiltonian **ii)** Hamiltonian but not Eulerian

c) Show that K_n has a Hamilton circuit whenever $n \geq 3$. **3**

8.a) Illustrate how Kruskal's algorithm and Prim's algorithm are used to find a minimum spanning tree, using a weighted graph with at least eight vertices and 15 edges. **5**

b) How can the final exams at a university be scheduled (using graph coloring) so that no student has two exams at the same time? **4**

c) In which order an in-order and post-order traversal visit the vertices of the given ordered rooted tree. **3**



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